Trust the Card, Not the Resemblance

Muhan Chen | Team FP12 | Symposium A

Course instructor: Professor Luyao Zhang

KEY MESSAGE

On a misaligned card, a public forfeit of 5 makes (Trust, Keep) the unique equilibrium. A similarity label changes no payoff. Trust is higher on the forfeit rounds (26/28), but those rounds are mixed with order. Keep did not rise: 14/28 aligned, 15/28 with no forfeit, and 15/28 with the forfeit.

01 Question and foundations

Research question

When the payoff card is misaligned, what public rule makes Trust a Nash equilibrium even if a SAME/DIFFERENT label is on the screen?

Closest work: what is inherited, and what is ours

Bracht and Feltovich (2008) let the allocator place an escrow, chosen or imposed. This paper inherits that precommitment, not a chosen deposit. The forfeit of 5 is printed before either choice. Neither player posts it.

Fehr and Rockenbach (2003) find that a sanction showing a selfish intention can destroy cooperation, while one seen as fair does not. Our forfeit is printed on the card. The Decider does not choose it. We did not ask whether the pairs saw it as fair.

Giang, Bell, and Buchner (2012) found that a self-resemblant face did not raise cooperation. Our cue is a priority label, not a face, and Trust was not higher under SAME.

Strategic model

The supplier is the Promiser (Keep or Break). The buyer is the Decider (Trust or Verify). Both lock before either choice is shown. Forfeit of 5: (Trust, Keep), payoffs (6, 6). No forfeit: (Verify, Break), payoffs (2, 10). Nashpy agrees at b1380f7. The pairs do not show that people follow the card.

Three lenses

Game theory: (Trust, Keep) with the forfeit; (Verify, Break) without it.

Social choice: (2, 10) to (6, 6). The Decider gains 4, the Promiser loses 4, and the total stays 12.

Mechanism design: the label cannot waive the printed forfeit.

Auction boundary, not the result

Bids choose the Promiser only. On a misaligned Break the forfeit equals the second bid. Fixed 5 keeps (Trust, Keep). A second bid must exceed 4. Bids (8, 3) give (Verify, Break). At 4 there is no unique equilibrium. Not played.

Named contribution

Muhan Chen is the sole author. Decisions: the Promiser chooses; the forfeit is 5 because Break gains 4; the label stays outside the matrix; two people play. Checks: dominance, the three Nashpy results, and the seed-206 replay, 48, 24, 22, and 18.

2056 contribution

By 2056 the aim is a tested rule for when a copied style cannot replace a public card. Checked on 19 October from the official pages, statement unchanged.

02 Model and verified evidence

	Aligned		No forfeit		Forfeit of 5	
	Keep	Break	Keep	Break	Keep	Break
Trust	(6, 6)	(−4, 3)	(6, 6)	(−4, 10)	(6, 6)	(−4, 5)
Verify	(2, 6)	(2, 3)	(2, 6)	(2, 10)	(2, 6)	(2, 5)
	(Trust, Keep)		(Verify, Break)		(Trust, Keep)	

Inherited
Trust game and cheap talk.

Contribution
A printed forfeit. The label cannot waive it.

Evidence
Nashpy agrees. The pairs do not show that people follow the card.

Figure 1. Cells are (Decider, Promiser). Shaded cells are the pure Nash equilibrium.

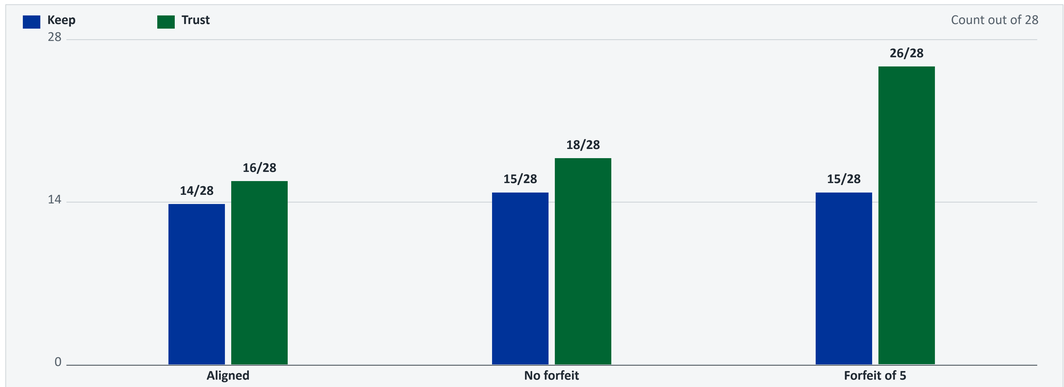


Figure 2. Forfeit cards are always rounds 5, 7, 11, and 12, so the forfeit is mixed with order. On those cards the expected payoff from Trust is about 1.36, below Verify's 2, yet Trust was 26/28. Exploratory. No p-value. Commit b1380f7.

Across pairs, Trust on the forfeit card rose in 5, was unchanged in 2, and fell in 0. Keep contrasts, forfeit minus no forfeit, are −1, −1, 0, +2, 0, +1, −1.
Aligned-card Trust is 16/28, not higher than Trust on the misaligned card with no forfeit, 18/28.

03 Impacts and roadmap

Figure 3. Targets 16.6 and 10.2 on the mechanism.

Proposed, not demonstrated

Mechanism
A collected forfeit of 5 moves the card from (Verify, Break) to (Trust, Keep). The label cannot waive it. The buyer decides. The supplier pays.

16.6
Develop effective, accountable and transparent institutions at all levels. Indicator: card, forfeit, and appeal visible before both choices lock.

10.2
Inclusion irrespective of economic or other status. Risk: who can pay a flat 5. Not indicator 10.2.1.

Evidence status: model result; exploratory classroom data; impact not demonstrated.

A written objection before collection is proposed, not built (**拟议，未实现**). It is not in the model or the Space. Charging the filer would add to the target 10.2 question of who can pay, so any such fee would be capped or waived.

Figure 4. One limit, three studies. **Keyword: Keep near 1/2**

Limit. Keep stays near one half where Keep pays more.

Economics
Question: why break when Keep pays more?
Method: noise, or a belief that collection fails.
Who: Zijie Wang and Lin Zhang, if they agree.
Test: can either story put Keep near one half?
Milestone: a note beside this matrix.

Computing
Question: randomize order, ask one payoff question, show collection?
Method: add the controls in this Space. Not a move to oTree.
Who: a software or HCI coauthor.
Test: a fresh run writes the pair contrasts.
Milestone: one commit after b1380f7.

Behavior
Question: is Keep higher once order, the check, and collection are split?
Method: at least 10 new pairs, not only code 206.
Who: a behavioral-science lab.
Test: the Keep contrast after the check and visible collection.
Milestone: one new session file. None exists.

Preset rule, not a test: at least 10 pairs who pass the payoff check and can see collection. If higher Keep is not more common than lower Keep, a tie included, replace the imposed 5 with an escrow the Promiser chooses.

Last week on the record

12 OCT — Ed duet with FP7
The claim taken in was the printed forfeit of 5. Zijie Wang selected the three Set A papers and noted that Keep stayed near one half. Lin Zhang worked through the duet. Confirmed response: the seven pairs do not show that people follow the card, and order is mixed with the forfeit. https://edstem.org/us/courses/102917/discussion/8391519?comment=19481056

14 OCT — self-challenge and FP7
The Promiser is paid the same whether anyone trusted, and a fine of 5 was treated as given. FP7 said their signal needs compliance. That question changed this roadmap: the next test is collection. Both equilibria total 12. Order was fixed, and collection was assumed.

21 OCT — final submission
Due at 23:00 Asia/Shanghai. This file is the final version, 20 Oct 2026, model commit b1380f7. The paper footer carries the same line.

Full acknowledgments

Prof. Luyao Zhang, formal reviewer: the PS1 final review of 25 September 2026 required a two-sided game and visible matrices. This project implements both.
I thank Prof. Kenneth S. (Ken) Rogerson for asking us, as Program Discussant of the 7 September 2026 workshop, to say what practical decision the result could inform, and to keep a classroom result separate from a real-world effect.
I thank Han Zhang for asking me to make the comparison in the paper easier to see, and to say who would be affected by the rule.
Zijie Wang and Lin Zhang (FP7) gave ideas in class on 12 and 14 October 2026. On 12 October, Zijie Wang selected Bracht and Feltovich (2008), Fehr and Rockenbach (2003), and Giang, Bell, and Buchner (2012), and noted that Keep stayed near one half. Lin Zhang worked through that duet with him.
Classmates whose teams I reviewed, or who reviewed this project, are invited to tell me if their names are missing. I will correct the paper and the poster.
Tencent's Shanghai office hosted the 4 September 2026 visit, including the technology exhibition and the AI chatbot session that framed a screen copying a user's priorities. The Shanghai Science and Technology Museum held the public AI exhibition on that visit. The whole COMSCI/ECON 206 classroom community produced the seven paired sessions and the 12 October duet.
Data: seven session files in data/. Software: Python, Nashpy, JavaScript, Colab, GitHub, and Hugging Face. Adapted materials: the PS1 seed-206 replay, and the trust-game and cheap-talk setting. AI assistance: Cursor and Claude, paper Appendix A.2.

References and open materials

Bracht and Feltovich 2008. DOI 10.1007/s00182-007-0092-3. Fehr and Rockenbach 2003. DOI 10.1038/nature01474. Giang, Bell, and Buchner 2012. DOI 10.1371/journal.pone.0047809.

GitHub **Hugging Face**

GitHub https://github.com/dku-comsci-econ206-Autumn2026/PS2-FP12-Muhan-Chen
Colab https://colab.research.google.com/github/dku-comsci-econ206-Autumn2026/PS2-FP12-Muhan-Chen/blob/b1380f7/PS2_trust_forfeit.ipynb
Hugging Face https://huggingface.co/spaces/dku-comsci-econ206-2026/Trust_Forfeit
Poster in the README https://github.com/dku-comsci-econ206-Autumn2026/PS2-FP12-Muhan-Chen/blob/main/README.md